

Math 2551 Worksheet: Triple Integrals in Cylindrical & Spherical Coordinates

During studio, do the set up portion of these integrals only. You can practice doing the evaluation after studio. Both the correct triple integral and correct final answer are given in the answers for this worksheet.

1. Convert the integral $\int_{-1}^1 \int_0^{\sqrt{1-y^2}} \int_0^x (x^2 + y^2) \, dz \, dx \, dy$ into an integral in cylindrical coordinates, and evaluate the integral.
2. Suppose $a \geq 0$. Find the volume of the region cut from the solid sphere $\rho \leq a$ by the half-planes $\theta = 0$ and $\theta = \pi/6$ in the first octant.
3. Find the volume of the solid that lies within the sphere $x^2 + y^2 + z^2 = 4$, above the xy -plane, and below the cone $z = \sqrt{x^2 + y^2}$.
4. Find the volume of the region bounded above by the paraboloid $z = 9 - x^2 - y^2$, below by the xy -plane, and lying *outside* the cylinder $x^2 + y^2 = 1$.
5. Find the volume of the solid that is between the spheres $\rho = \sqrt{2}$ and $\rho = 2$, but outside of the circular cylinder $x^2 + y^2 = 1$. It will be helpful to draw a cross-section in a plane $\theta = c$ for this problem and to use symmetry.
6. Let D be the right circular cylinder whose base is the circle $r = 2 \sin \theta$ in the xy -plane and whose top lies in plane $z = 4 - y$. Recall that $r = 2 \sin \theta$ describes a circle centered at $(0, 1)$ with radius 1 in the xy -plane. Using cylindrical coordinates,
 - (a) find the volume of the region D .
 - (b) find the $\bar{x}$ component of the centroid of the region (hint: use symmetry).